

Taowen Liu

Phone: +86 13533593419 Email: h98798149@gmail.com

EDUCATION

Harbin Institute of Technology, Shenzhen

|Shenzhen, China

B.Eng. in Automation

Aug. 2022 – Jun. 2026

- Average Mark: 90.9/100
- Weighted GPA: 3.703/4.0 (Rank:36/264)

Relevant Coursework:

- C++ Language Programming (94), High-level Language Programming(87)
- Electric Circuit(91), Fundamentals of Analog Electronics, Digital electronic technology foundation
- Pattern Recognition (92), Digital Image Processing(89), System modeling and Simulation(88), Automatic Control Theory(94), Automatic Control Practice(86)

English Skills:

- IELTS: 7.5 (L: 8.5 R: 8.5 W: 7 S: 6.5)
- CET-6: 621

INTERNSHIP

Auto-Tuning Syetem for Battery Configuration | Shenzhen Yiwei Microelectronics Technology Co., Ltd.

Enterprise Internship; *Advisor: Dong Li (CTO)*

Jan. 2025 – May. 2025

- Developed software modules for the **Auto-Tuning System (ATS)** used in **battery parameter configuration**.
- Designed auto-tuning algorithms based on **negative feedback** and **open-loop correction** for **curve smoothing** and **parameter calibration**.
- Successfully ensured compliance with **UI/U2 testing standards** set by battery manufacturers, and **deployed** the algorithms within the company's **ATS software platform**.

Software Testing Intern

| Shenzhen Jijia Innovation Technology Co., Ltd.

Enterprise Internship; *Advisor: Guangzhen Lin (Tutor)*

Aug. 2025 – Nov.2025

- Participate in **embedded software testing** for IoT and smart camera products, including basic **functional regression testing**, as well as system-level testing at the **finished product** manufacturer and board-level testing at the **PCBA** manufacturer.
- Design and execute **test cases**, report **bugs**, and collaborate with developers to improve product stability.
- Assist in **automating regression testing scripts**, improving testing efficiency and coverage.

RESEARCH

Motion Compensation of Stewart Platform Based on AprilTag

Aug. 2024 – Jun.2025

| Harbin Institute of Technology, Shenzhen

Research Intern; *Advisor: Dr. Peng Xu*

- Assisted in designing a custom six-degree-of-freedom parallel Stewart platform based on the Gough-Stewart mechanism.
- Proposed the use of the **April Tag method** to monitor the **six-degree-of-freedom pose** of the camera relative to fixed tags, with the data fed back to the motion compensation control module.

Prognostication of Therapeutic Efficacy in Epilepsy Based on Responsive Neurostimulation

Aug. 2025 – Jun.2026

| Harbin Institute of Technology, Shenzhen

Undergraduate Thesis; *Advisor: Assoc. Prof. Di Wu*

- Developed a **dual-branch LSTM-Attention model** integrating raw and differential EEG features to capture both **signal amplitude and temporal trends**.
- Designed a time-frequency **EEG feature extraction** and week-level temporal **aggregation**

pipeline for neural signal modeling.

- Proposed a patient-level labeling strategy using longitudinal **seizure burden variables** for more **clinically reliable outcome evaluation**.
- Performed **interpretable analysis** using **feature importance** and **attention visualization** to identify clinically relevant biomarkers and key temporal windows.

PROJECTS

Dual-CCD Line-Following Smart Car

| Harbin Institute of Technology, Shenzhen

Advisor: Prof. Ya ming Ge

Oct. 2023 – Jul. 2024

- Developed a **dual-CCD line-following smart car** based on the **STC32 microcontroller**.
- Implemented **path prediction and error feedback control** using dual CCD sensors for **complex track recognition**.
- Designed a **PID-based feedback** controller for stable trajectory tracking.
- Independently designed the **main control PCB and motor driver board**.

Intelligent Safe Driving Monitoring System

| Harbin Institute of Technology, Shenzhen

Advisor: Prof. Jin xing Li

Sep. 2026 – May. 2026

- Developed a **lane detection and lane departure** warning system based on the **UFLD model**.
- Built a **Driver Monitoring System** using **YOLOv8** to detect fatigue and distracted driving behaviors.
- Deployed deep learning models on Android devices via **OpenBot** for real-time **edge inference**.
- Designed a web/mobile monitoring interface integrated with the **Amap API** for real-time **vehicle location and weather visualization**.

HONORS & AWARDS

- Second Prize in the South China Division of the 19th National University Students Intelligent Car Race 2024
- Third Prize in the Provincial Division of the China Undergraduate Mathematical Contest in Modeling (CUMCM) 2024
- University Second Prize for First-Year Annual Project 2022

SKILLS

- **Programming:** Python, C/C++, MATLAB
 - **Robotics & Embedded Systems:** ROS, STM32/STC32, Arduino, PID Control, Motion Control, Sensor Integration
 - **AI & Computer Vision:** PyTorch, OpenCV, YOLO, Deep Learning models, Attention Mechanisms, RAG
 - **Tools & Platforms:** Ubuntu/Linux, Git, OpenBot, SolidWorks
-